```
User.where("last_name ILIKE 'lname123%'").estimated_count
```

The estimated_count() method is defined in the "fast_counts" gem. The estimated count is a bit off from the real count value but is tremendously faster to get.

Finally, if you need to count distinct values, the PostgreSQL wiki documents some tricks on how to do this quickly (See: *Selecting Distinct Values*[22]). Fast distinct counts are being worked on as an addition to the fast_count gem at publication time.

Another advanced technique is to use the postgresql-hll extension. The HyperLogLog (HLL) concept is used to estimate the number of distinct values in a large set. The HyperLogLog is described as a *cardinality estimator*.

After installing and enabling the PostgreSQL extension, the HyperLogLog functionality may be used via the active-hll[23] gem. Try configuring it with Rideshare and installing the gem if you'd like to explore it.

If estimates or these tricks won't work, you may need to manage your own tally of rows at Insert time. The general technique is similar to how the Active Record Counter Cache works.

When reporting multiple COUNT values a CASE statement may be used. The CASE statement is used to set up different integer values and then sum the values.

An alternative to using SUM with CASE and WHEN is the lesser-known FILTER keyword, which can also be used to display multiple counts at once. The post *The FILTER clause in Postgres 9.4*[24] shows how to use FILTER for "scoped aggregate functions."

Try this out for Rideshare. Run this statement from psql:

sql/count_with_filter.sql

```
SELECT
  COUNT(*) FILTER (WHERE type = 'Rider') AS rider_count,
  COUNT(*) FILTER (WHERE type = 'Driver') AS driver_count
FROM users;
```

The query above displays the counts of Riders and Drivers. Although it scans the table, it does only one scan.

You've now seen some ways to improve the performance for count operations. Next you'll look at some tools that will help you with your database design.

22. https://wiki.postgresql.org/wiki/Loose_indexscan
23. https://github.com/ankane/active_hll
24. https://medium.com/little-programming-joys/the-filter-clause-in-postgres-9-4-3dd327d3c852